

Quiz (Carolina Reaper)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{2}}$.

- (A) $\frac{\sqrt{2}}{2}$
- (B) $\frac{1}{2\sqrt{2}}$
- (C) $\frac{2}{\sqrt{2}}$
- (D) $\frac{\sqrt{2}}{\sqrt{2}}$
- (E) $\frac{1}{2}$

Q2. Simplify $\frac{1}{\sqrt{5+1}}$.

- (A) $\frac{\sqrt{5}-1}{4}$
- (B) $\frac{\sqrt{5}+1}{4}$
- (C) $\frac{1}{\sqrt{5}}$
- (D) $\frac{\sqrt{5}-1}{5}$
- (E) $\frac{1}{5}$

Q3. Rationalize the denominator of $\frac{2}{\sqrt{7}-1}$.

- (A) $\frac{2(\sqrt{7}+1)}{6}$
- (B) $\frac{2(\sqrt{7}-1)}{6}$
- (C) $\frac{2}{\sqrt{7}}$
- (D) $\frac{\sqrt{7}+1}{2}$
- (E) $\frac{\sqrt{7}-1}{2}$

Q4. Simplify $\frac{3}{\sqrt{11}-2}$.

- (A) $\frac{3(\sqrt{11}+2)}{3}$
- (B) $\frac{3(\sqrt{11}-2)}{3}$
- (C) $\frac{3}{\sqrt{11}}$
- (D) $\frac{\sqrt{11}+2}{3}$
- (E) $\frac{\sqrt{11}-2}{3}$

Q5. Rationalize the denominator of $\frac{1}{\sqrt{3}+2}$.

- (A) $\frac{\sqrt{3}-2}{-1}$
- (B) $\frac{\sqrt{3}+2}{-1}$
- (C) $\frac{\sqrt{3}-2}{1}$
- (D) $\frac{1}{\sqrt{3}}$
- (E) $\frac{2}{\sqrt{3}}$

Q6. Simplify $\frac{2}{\sqrt{13}+3}$.

- (A) $\frac{2(\sqrt{13}-3)}{4}$
- (B) $\frac{2(\sqrt{13}+3)}{4}$
- (C) $\frac{\sqrt{13}-3}{2}$
- (D) $\frac{\sqrt{13}+3}{2}$
- (E) $\frac{2}{\sqrt{13}}$

Q7. Rationalize the denominator of $\frac{4}{\sqrt{17}-1}$.

- (A) $\frac{4(\sqrt{17}+1)}{16}$
- (B) $\frac{4(\sqrt{17}-1)}{16}$
- (C) $\frac{4}{\sqrt{17}}$
- (D) $\frac{\sqrt{17}+1}{4}$
- (E) $\frac{\sqrt{17}-1}{4}$

Q8. Simplify $\frac{5}{\sqrt{23}+4}$.

- (A) $\frac{5(\sqrt{23}-4)}{7}$
- (B) $\frac{5(\sqrt{23}+4)}{7}$
- (C) $\frac{\sqrt{23}-4}{5}$
- (D) $\frac{\sqrt{23}+4}{5}$
- (E) $\frac{5}{\sqrt{23}}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{3}{\sqrt{2}+\sqrt{5}}$ and simplify.

Q10. Rationalize the denominator of $\frac{2}{\sqrt{3}-\sqrt{2}}$ and simplify.

Chef's Tips

- Double-check calculations
- Show all work
- Use correct notation